

Jacob C. Stallings

Nashville, TN, USA | stallingsaudio@gmail.com

Education

MS, Audio Engineering | Belmont University

May 2023

- Coursework: Deep Learning & AI Ethics (PyTorch), Audio Software Development (C++, JUCE), Audio Signal Processing (Matlab), Audio Mastering & Applied Critical Listening
- Projects: Digital Audio Effect Modeling w/ PyTorch, Distortion VST/AU Plugin w/ JUCE, Applied Mastering, Measurement of Head-Related Transfer Functions

BS, Computer Science | Vanderbilt University

December 2020

- Coursework: Cloud Computing (Virtualization, Networking, Docker, Kubernetes, AWS), Artificial Intelligence, Computer Architecture, Operating Systems (Linux, C), Algorithms (Performance Optimization, Big-O Analysis, Graph Algorithms), Statistics, Linear Algebra, Spanish

Technical Skills

Languages: Proficient: Python, C++ (JUCE), Java, JavaScript (Node.js / React), SQL; Working: OpenGL, HTML/CSS (Tailwind), Matlab

Other Skills & Tools: PyTorch, REST APIs, MongoDB, PostgreSQL, Git, SVN, Cursor / AI Agents, SSH, AWS (S3, Lambda, SQS, EC2, SES, CloudFormation, AppSync), GraphQL, Docker, Jupyter Notebooks

Work Experience

Genesis Studio

June 2024 - Present

Full-Stack Engineer, Artist Evolve

Remote (Nashville)

- Implemented social platform API integrations: Content scheduling, metrics & insights, auth flows
- Architected and implemented a multi-channel live notification system with GraphQL + AWS AppSync
- Implement and maintain UI elements: Interactive timeline view (React + Tailwind), Email HTML templates
- Designed and maintain CI/CD for multiple production Lambdas with environment-specific workflows (AWS)

Hack Audio, LLC

July 2024 - Present

Developer, Web & Audio UI (Part-time)

Remote (Nashville)

- Overhauled the public-facing website UI using WordPress and custom CSS, driving 1,000+ monthly views
- Developing a JUCE 8 + WebView-powered preset browser feature, with live registry hosted via WordPress

Bender Audio, LLC

Feb 2024 - Oct 2025

Lead Developer, ClipperX

Remote (Nashville)

- Implemented fully controllable professional audio effect in JUCE with real-time audio visualization
- Instrumented and optimized real-time waveform visualization with cross-platform OpenGL GPU shaders
- Communicated technical requirements with stakeholders and coordinated smooth handoff to dev team

Vitech Systems Group

Jan 2022 - Apr 2024

Backend Engineer, Disbursements Team

Remote (Nashville)

- Developed features and bug fixes for V3locity, a cloud-native, web-based insurance admin platform
- Technologies used: Java (Spring), Hibernate, PostgreSQL, REST, Node.js, TypeScript, React, Git, SVN
- Participated in Scrum ceremonies such as backlog refinement and code reviews

Other Highlights

- 10 years experience in music production (Ableton Live / Logic Pro), releases on labels such as Houstricke
- Attended the Generative AI Music Workshop hosted by The Sound of AI and Pompeu Fabra University (Music Technology Group) in Barcelona, Spain, December 2023
- Juni Learning: Taught 1-on-1 programming classes to students aged 11-17 in Python, Java, HTML/CSS/JS
- Studied one semester of undergraduate computer science at Carlos III University in Madrid, Spain